Exercise J - A

1. Let $-\frac{\pi}{6} < \theta < -\frac{\pi}{12}$. Suppose α_1 and β_1 are the roots of the equation $x^2 - 2x\sec\theta + 1 = 0$ and α_2 and β_2 are the roots of the equation $x^2 + 2x\tan\theta - 1 = 0$. If $\alpha_1 > \beta_1$ and $\alpha_2 > \beta_2$, then $\alpha_1 + \beta_2$ equals : **[JEEAdv-2016]**

माना कि $-\frac{\pi}{6} < \theta < -\frac{\pi}{12}$ है। मान लिजिये कि α_1 और β_1 समीकरण $x^2 - 2x\sec\theta + 1 = 0$ के मूल (roots) हैं और α_2 और β_2 समीकरण $x^2 + 2x\tan\theta - 1 = 0$ के मूल हैं। यदि $\alpha_1 > \beta_1$ और $\alpha_2 > \beta_2$ हैं, जब $\alpha_1 + \beta_2$ का मान है : **[JEEAdv-2016]**

(A) $2(\sec\theta - \tan\theta)$ (B) $2\sec\theta$ (C) $-2\tan\theta$ (D) 0

Ans. C

Sol. Roots of the equation $x^2 - 2x\sec\theta + 1 = 0$ are $\sec\theta - \tan\theta$, $\sec\theta + \tan\theta$

$\alpha_1 = \sec\theta - \tan\theta$, $\beta_1 = \sec\theta + \tan\theta$ $(\because \alpha_1 > \beta_1)$

Roots of the equation $x^2 + 2x\tan\theta - 1 = 0$ are $-\tan\theta + \sec\theta$, $-\tan\theta - \sec\theta$

$\alpha_2 = \sec\theta - \tan\theta$, $\beta_2 = -\sec\theta - \tan\theta$ $(\because \alpha_2 > \beta_2)$

Hence $\alpha_1 + \beta_2 = -2\tan\theta$

2. The value of $\sum_{k=1}^{13} \frac{1}{\sin\left(\frac{\pi}{4} + \frac{(k-1)\pi}{6}\right)\sin\left(\frac{\pi}{4} + \frac{k\pi}{6}\right)}$ is equal to : **[JEEAdv-2016]**

$\sum_{k=1}^{13} \frac{1}{\sin\left(\frac{\pi}{4} + \frac{(k-1)\pi}{6}\right)\sin\left(\frac{\pi}{4} + \frac{k\pi}{6}\right)}$ का मान है : **[JEEAdv-2016]**

(A) $3 - \sqrt{3}$ (B) $2\left(3 - \sqrt{3}\right)$ (C) $2\left(\sqrt{3} - 1\right)$ (D) $2\left(2 + \sqrt{3}\right)$

Ans. C

Sol. $$\sum_{k=1}^{13} \frac{\sin\left[\left(\frac{\pi}{4} + \frac{k\pi}{6}\right) - \left(\frac{\pi}{4} + (k-1)\frac{\pi}{6}\right)\right]}{\sin\frac{\pi}{6}\left(\sin\left(\frac{\pi}{4} + \frac{k\pi}{6}\right)\sin\left(\frac{\pi}{4} + (k-1)\frac{\pi}{6}\right)\right)} = 2\sum_{k=1}^{13}\left(\cot\left(\frac{\pi}{4} + (k-1)\frac{\pi}{6}\right) - \cot\left(\frac{\pi}{4} + \frac{k\pi}{6}\right)\right)$$

$$= 2\left(\cot\frac{\pi}{4} - \cot\left(\frac{\pi}{4} + \frac{13\pi}{6}\right)\right) = 2\left(1 - \cot\left(\frac{29\pi}{12}\right)\right) = 2\left(1 - \cot\left(\frac{5\pi}{12}\right)\right) = 2(1 - (2 - \sqrt{3})) = 2(-1 + \sqrt{3})$$

$$= 2\left(\sqrt{3} - 1\right)$$

3. Let α and β be non-zero real numbers such that $2(\cos\beta - \cos\alpha) + \cos\alpha\cos\beta = 1$. Then which of the following is/are true : **[JEEAdv-2017]**

माना कि α तथा β इस प्रकार की अशून्य वास्तविक संख्यायें हैं, कि $2(\cos\beta - \cos\alpha) + \cos\alpha\cos\beta = 1$. तब निम्न में से कौनसा/कौनसे कथन सत्य है/हैं : **[JEEAdv-2017]**

(A) $\tan\left(\frac{\alpha}{2}\right) + \sqrt{3}\tan\left(\frac{\beta}{2}\right) = 0$

(B) $\sqrt{3}\tan\left(\frac{\alpha}{2}\right) - \tan\left(\frac{\beta}{2}\right) = 0$

(C) $\tan\left(\frac{\alpha}{2}\right) - \sqrt{3}\tan\left(\frac{\beta}{2}\right) = 0$

(D) $\sqrt{3}\tan\left(\frac{\alpha}{2}\right) + \tan\left(\frac{\beta}{2}\right) = 0$

Ans. AC

Sol. $\cos\alpha = \left(\frac{1-a}{1+a}\right); \qquad a = \tan^2\frac{\alpha}{2}$

$\cos\beta = \left(\frac{1-b}{1+b}\right); \qquad b = \tan^2\frac{\beta}{2}$

$$2\left(\left(\frac{1-b}{1+b}\right) - \left(\frac{1-a}{1+a}\right)\right) + \left(\left(\frac{1-a}{1+a}\right)\left(\frac{1-b}{1+b}\right)\right) = 1$$

$\Rightarrow$ $2((1 - b)(1 + a) - (1 - a)(1 + b)) + (1 - a)(1 - b) = (1 + a)(1 + b)$
$\Rightarrow 2(1 + a - b - ab - (1 + b - a - ab)) + 1 - a - b + ab = 1 + a + b + ab$
$\Rightarrow 4(a - b) = 2(a + b)$
$\Rightarrow 2a - 2b = a + b$
$\Rightarrow a = 3b$

$$\tan^2\frac{\alpha}{2} = 3\tan^2\frac{\beta}{2}$$

$$\tan\frac{\alpha}{2} = \pm\sqrt{3}\tan\left(\frac{\beta}{2}\right)$$

4. Let $f:[0,2] \to R$ be the function defined by $f(x) = (3 - \sin(2\pi x))\sin(\pi x - \frac{\pi}{4}) - \sin(3\pi x + \frac{\pi}{4})$. If $\alpha, \beta \in [0, 2]$ are such that $\{x \in [0, 2]: f(x) \geq 0\} = [\alpha, \beta]$, then the value of $\beta - \alpha$ is. **[JEEAdv-2020]**

माना कि फलन $f:[0,2] \to R$ को $f(x) = (3 - \sin(2\pi x)) \sin(\pi x - \frac{\pi}{4}) - \sin(3\pi x + \frac{\pi}{4})$ द्वारा परिभाषित किया जाता है। यदि $\alpha, \beta \in [0,2]$ इस प्रकार से हैं कि $\{x \in [0, 2]: f(x) \geq 0\} = [\alpha, \beta]$, तब $\beta - \alpha$ का मान है। **[JEEAdv-2020]**

Ans. 1

Sol. $f(x) = (3 - \sin 2\pi x) \sin\left(\pi x - \frac{\pi}{4}\right) - \sin\left(3\pi x + \frac{\pi}{4}\right)$

Let $\pi x - \frac{\pi}{4} = \pi y \Rightarrow x = y + \frac{1}{4}$

$f(x) = \left(3 - \sin 2\pi\left(y + \frac{1}{4}\right)\right) \sin y\pi - \sin\left(3\pi\left(y + \frac{1}{4}\right) + \frac{\pi}{4}\right)$

$= \left(3 - \sin\left(2\pi y + \frac{\pi}{2}\right)\right) \sin y\pi - \sin\left(3\pi y + \frac{3\pi}{4} + \frac{\pi}{4}\right)$

$= (3 - \cos 2\pi y) \sin\pi y + \sin(3\pi y)$

$= (3 - (1 - 2\sin^2 \pi y)) \sin \pi y + 3 \sin \pi y - 4 \sin^3 \pi y$

$= 5 \sin \pi y + 2 \sin^3 \pi y = \sin \pi y (5 + 2\sin^2 \pi y)$

Since $f(x) \geq 0 \sin \pi y \geq 0$

$\Rightarrow \sin \pi \left(x - \frac{1}{4}\right) \geq 0 \Rightarrow \pi \left(x - \frac{\pi}{4}\right) \in [2n\pi, (2n + 1)\pi]$

$x \in [0, 2] \Rightarrow \pi x - \frac{\pi}{4} \in \left[-\frac{\pi}{4}, \frac{7\pi}{4}\right]$

From both of these,

$\pi x - \frac{\pi}{4} \in [0, \pi]$

$\Rightarrow x \in \left[\frac{1}{4}, \frac{5}{4}\right]$

So, $\alpha = \frac{1}{4}, \beta = \frac{5}{4}$

$\beta - \alpha = \frac{5}{4} - \frac{1}{4} = 1$

5. Let α and β be real numbers such that $-\frac{\pi}{4} < \beta < 0 < \alpha < \frac{\pi}{4}$. If $\sin(\alpha + \beta) = \frac{1}{3}$ and $\cos(\alpha - \beta) = \frac{2}{3}$, then the greatest integer less than or equal to $\left(\frac{\sin\alpha}{\cos\beta} + \frac{\cos\beta}{\sin\alpha} + \frac{\cos\alpha}{\sin\beta} + \frac{\sin\beta}{\cos\alpha}\right)^2$ is _____.

[JEEAdv-2022]

माना कि α एवं β ऐसी वास्तविक संख्यायें हैं कि $-\frac{\pi}{4}<\beta<0<\alpha<\frac{\pi}{4}$ है। यदि $\sin(\alpha+\beta)=\frac{1}{3}$ एवं $\cos(\alpha-\beta)=\frac{2}{3}$ है, तब

$\left(\frac{\sin\alpha}{\cos\beta}+\frac{\cos\beta}{\sin\alpha}+\frac{\cos\alpha}{\sin\beta}+\frac{\sin\beta}{\cos\alpha}\right)^2$ से कम या बराबर महत्तम पूर्णांक _____ है। **[JEEAdv-2022]**

Ans. 1

Sol. $\left(\frac{\sin\alpha}{\cos\beta}+\frac{\cos\alpha}{\sin\beta}+\frac{\cos\beta}{\sin\alpha}+\frac{\sin\beta}{\cos\alpha}\right)^2$

$=\left(\frac{\cos(\alpha-\beta)}{\sin\beta\cos\beta}+\frac{\cos(\alpha-\beta)}{\sin\alpha\cdot\cos\alpha}\right)^2$

$=\left(\frac{4}{3}\left\{\frac{1}{\sin2\beta}+\frac{1}{\sin2\alpha}\right\}\right)^2$

$=\frac{16}{9}\left(\frac{2\sin(\alpha+\beta)\cdot\cos(\alpha-\beta)}{\sin2\alpha\cdot\sin2\beta}\right)^2$

$=\frac{16}{9}\left(\frac{4\cdot\frac{1}{3}\cdot\frac{2}{3}}{\cos(2\alpha-2\beta)-\cos(2\alpha+2\beta)}\right)^2$

$=\frac{16}{9}\left(\frac{\frac{8}{9}}{2\cos^2(\alpha-\beta)-1-1+2\sin^2(\alpha+\beta)}\right)^2$

$=\frac{16}{9}\left(\frac{\frac{8}{9}}{\frac{8}{9}-2+\frac{2}{9}}\right)^2 \qquad =\frac{16}{9}$

6. Let $\frac{\pi}{2}<x<\pi$ be such that $\cot x=\frac{-5}{\sqrt{11}}$. Then

$\left(\sin\frac{11x}{2}\right)(\sin6x-\cos6x)+\left(\cos\frac{11x}{2}\right)(\sin6x+\cos6x)$ is equal to : **[JEE(Adv.)2024]**

माना कि $\frac{\pi}{2} < x < \pi$ इस प्रकार है कि $\cot x = \frac{-5}{\sqrt{11}}$ है। तब

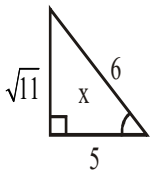
$\left(\sin\frac{11x}{2}\right)(\sin 6x - \cos 6x) + \left(\cos\frac{11x}{2}\right)(\sin 6x + \cos 6x)$ बराबर है : : **[JEE(Adv.)2024]**

(A) $\frac{\sqrt{11}-1}{2\sqrt{3}}$ (B) $\frac{\sqrt{11}+1}{2\sqrt{3}}$ (C) $\frac{\sqrt{11}+1}{3\sqrt{2}}$ (D) $\frac{\sqrt{11}-1}{3\sqrt{2}}$

Ans. B

Sol. $\frac{\pi}{2} < x < \pi$

$\cot x = -\frac{5}{\sqrt{11}}$



$\left(\sin\frac{11x}{2}\right)(\sin 6x - \cos 6x) + \left(\cos\frac{11x}{2}\right)(\sin 6x + \cos 6x)$

$\left(\cos 6x\cos 11\frac{x}{2} + \sin 6x\sin 11\frac{x}{2}\right) + \left(\sin 6x\cos 11\frac{x}{2} - \cos 6x\sin 11\frac{x}{2}\right)$

$\Rightarrow \cos\left(6x - 11\frac{x}{2}\right) + \sin\left(6x - 11\frac{x}{2}\right) = \cos\frac{x}{2} + \sin\frac{x}{2}$

$\Rightarrow \sqrt{(1+\sin x)^2}$

$= \sqrt{\left(1+\frac{\sqrt{11}}{6}\right)^2}$

$\Rightarrow \sqrt{\left(1+\frac{\sqrt{11}}{6}\right)^2} = \sqrt{\frac{12+2\sqrt{11}}{12}}$

$= \frac{\sqrt{11}+1}{2\sqrt{3}}$

7. Let $\alpha = \frac{1}{\sin 60^\circ \sin 61^\circ} + \frac{1}{\sin 62^\circ \sin 63^\circ} + \cdots + \frac{1}{\sin 118^\circ \sin 119^\circ}$. Then the value of $\left(\frac{\text{cosec}\,1^\circ}{\alpha}\right)^2$ is _____.

[JEE(Adv.)2025]

मान लीजिए कि $\alpha = \frac{1}{\sin 60^\circ \sin 61^\circ} + \frac{1}{\sin 62^\circ \sin 63^\circ} + \cdots + \frac{1}{\sin 118^\circ \sin 119^\circ}$ है। तब $\left(\frac{\text{cosec}\,1^\circ}{\alpha}\right)^2$ का मान _____ है।

[JEE(Adv.)2025]

Ans. 3

Sol. $(\alpha)\ \sin 1^\circ = \frac{\sin(61^\circ - 60^\circ)}{\sin 60^\circ \sin 61^\circ} + \frac{\sin(63^\circ - 62^\circ)}{\sin 63^\circ \sin 62^\circ} + \ldots\ldots + \frac{\sin(119^\circ - 118^\circ)}{\sin 118^\circ \sin 119^\circ}$

$= (\cot 60° - \cot 61°) + (\cot 62° - \cot 63°) + \ldots\ldots\ldots + (\cot 118° - \cot 119°)$

$119° = 180° - 61° \quad \Rightarrow \cot 119° = -\cot 61°$

$\Rightarrow (\cot 60° + \cot 119°) + (-\cot 118° + \cot 117°) + \ldots\ldots\ldots\ldots + (\cot 118° - \cot 119°)$

$\Rightarrow \alpha = \frac{\cot 60^\circ}{\sin 10^\circ}$

$\text{RSH} \left(\frac{\text{cosec}\,1°}{\cot 60°} \sin 1°\right)^2 = 3$

8. Let $\alpha = \left(1 - 2\cos\left(\frac{\pi}{11}\right)\right)\left(1 - 2\cos\left(\frac{3\pi}{11}\right)\right)\left(1 - 2\cos\left(\frac{9\pi}{11}\right)\right)\left(1 - 2\cos\left(\frac{27\pi}{11}\right)\right)\left(1 - 2\cos\left(\frac{81\pi}{11}\right)\right)$. Then the value of $5 - \alpha^2$ is _____.

[JEE(Adv.)2026]

मान लीजिए कि $\alpha = \left(1 - 2\cos\left(\frac{\pi}{11}\right)\right)\left(1 - 2\cos\left(\frac{3\pi}{11}\right)\right)\left(1 - 2\cos\left(\frac{9\pi}{11}\right)\right)\left(1 - 2\cos\left(\frac{27\pi}{11}\right)\right)\left(1 - 2\cos\left(\frac{81\pi}{11}\right)\right)$ है। तब $5 - \alpha^2$ का मान _____ है।

[JEE(Adv.)2026]

Ans. 4

Sol. $\alpha = \left(1 - 2\cos\frac{\pi}{11}\right)\left(1 - 2\cos\frac{3\pi}{11}\right)\left(1 - 2\cos\frac{9\pi}{11}\right)\left(1 - 2\cos\frac{27\pi}{11}\right)\left(1 - 2\cos\frac{81\pi}{11}\right)$

Let $\theta = \frac{\pi}{11}$

$\alpha = (1 - 2\cos\theta)(1 - 2\cos 3\theta)(1 - 2\cos 9\theta)(1 - 2\cos 27\theta)(1 - 2\cos 81\theta)$

Now $1-2\cos\theta$

$$1-2\left\{2\cos^2\frac{\theta}{2}-1\right\}$$

$$3-4\cos^2\theta/2$$

$$\frac{\left(3-4\cos^2\theta/2\right)\cdot\cos\theta/2}{\cos\theta/2}$$

$$\Rightarrow -\frac{\cos\frac{3\theta}{2}}{\cos\frac{\theta}{2}}$$

$$\therefore\ \alpha=\left(-\frac{\cos\frac{3\theta}{2}}{\cos\frac{\theta}{2}}\right)\left(-\frac{\cos\frac{9\theta}{2}}{\cos\frac{3\theta}{2}}\right).....\left(-\frac{\cos\frac{243\theta}{2}}{\cos\frac{81\theta}{2}}\right)\quad=-\frac{\cos\frac{243\theta}{2}}{\cos\frac{\theta}{2}}$$

$$\text{Now }\theta=\frac{\pi}{11}\Rightarrow\ \frac{\theta}{2}\ =\frac{\pi}{22}\Rightarrow\quad 243\frac{\theta}{2}\ =\frac{\pi}{22}\times 243\qquad =11\pi+\frac{\pi}{22}$$

$$\cos 243\frac{\theta}{2}\ =-\cos\frac{\pi}{22}\ =-\cos\frac{\theta}{2}$$

$$\therefore\ \alpha=1\qquad\Rightarrow\qquad 5-\alpha^2=4$$